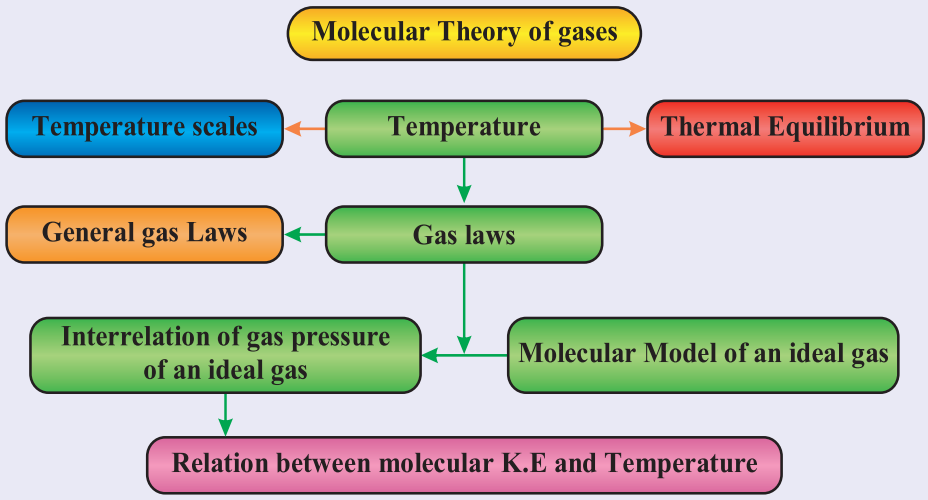
CONCEPT MAP



EXERCISE

Section (A): **Multiple Choice Questions (MCQs)**
Choose the correct answer:

1. The relationship between temperature and average kinetic energy of particles in a gas is:
   (a) temperature is inversely proportional to average kinetic energy
   (b) temperature is directly proportional to average kinetic energy
   (c) temperature is independent of average kinetic energy
   (d) temperature is proportional to the square of average kinetic energy

2. Standard conditions of temperature and pressure (STP) refer to a gas at:
   (a) 0°C and 1 atm (b) 20°C and 1 atm
   (c) 25°C and 1 atm (d) 30°C and 101.3 kPa (1 atm)

3. If the temperature is kept constant and the volume of a gas is doubled, then pressure of a gas is:
   (a) Reduced to $\frac{1}{2}$ of the original value. (b) Doubled
   (c) Reduced to $\frac{1}{4}$ of the original value (d) Quadrupled

4. The Avogadro's number is the number of molecules in:
   (a) One mole of a substance (b) One kg of a substance
   (c) One m^3 of a gas (d) One kilogram of hydrogen gas

5. Mean translational K.E. per molecule of an ideal at temperature T is:
   (a) $\frac{3}{2}kT$ (b) $\frac{1}{2}kT$ (c) $\frac{2}{3}kT$ (d) kT^4

6. The normal human body temperature is:
   (a) 98.6°F (37°C) (b) 99.6°F (37.4°C)
   (c) 100.4°F (38°C) (d) 101°F (38.3°C)

7. The pressure P, the density ρ and the average speed of molecules of an ideal gas are related by the equation.
   (a) $P = \frac{2}{3} m\ell^2$ (b) $P = \frac{1}{3} m\ell^2$ (c) $P = \frac{1}{3}\rho\overline{V^2}$ (d) $P = \frac{2}{3}\ell^2$

8. In air at S.T.P, the average speed of the:
   (a) Oxygen molecules is greater than Nitrogen molecules
   (b) Nitrogen molecules is greater than Oxygen molecules
   (c) Oxygen molecules is approximately equal to Nitrogen molecules
   (d) Helium atoms is greater than both Oxygen and Nitrogen molecules

9. If the absolute temperature of a gas is increased 3 times, the rms velocity of the molecules will be:
   (a) 3 times (b) 9 times (c) $\sqrt{3}$ times (d) $\frac{1}{3}$ times

10. A gas is enclosed in an isolated container which is placed on a fast-moving train uniformly. The temperature of the gas:
    (a) Increases due to the motion of the train
    (b) Decreases due to the motion of the train
    (c) Remains constant
    (d) Fluctuates, depending on the train's speed and direction

Section (B): CRQs (Short Answered Questions):

15.1. Why the earth is not in thermal equilibrium with the sun?
15.2. Describe the relationship between temperature and kinetic energy of molecules.
15.3. It is observed that when mercury in glass thermometer is put in a flame, the column of mercury first descends and then rises. Explain.
15.4. What is standard temperature, pressure?
15.5. A thermometer is placed in direct sun light. What will it read the temperature?
15.6. The pressure in a gas cylinder containing hydrogen will leak more quickly than if it is containing oxygen. Why?
15.7. When a sealed thermos bottle full of coffee is shaken, what are the changes occur?



Section (A): Multiple Choice Questions (MCQs)

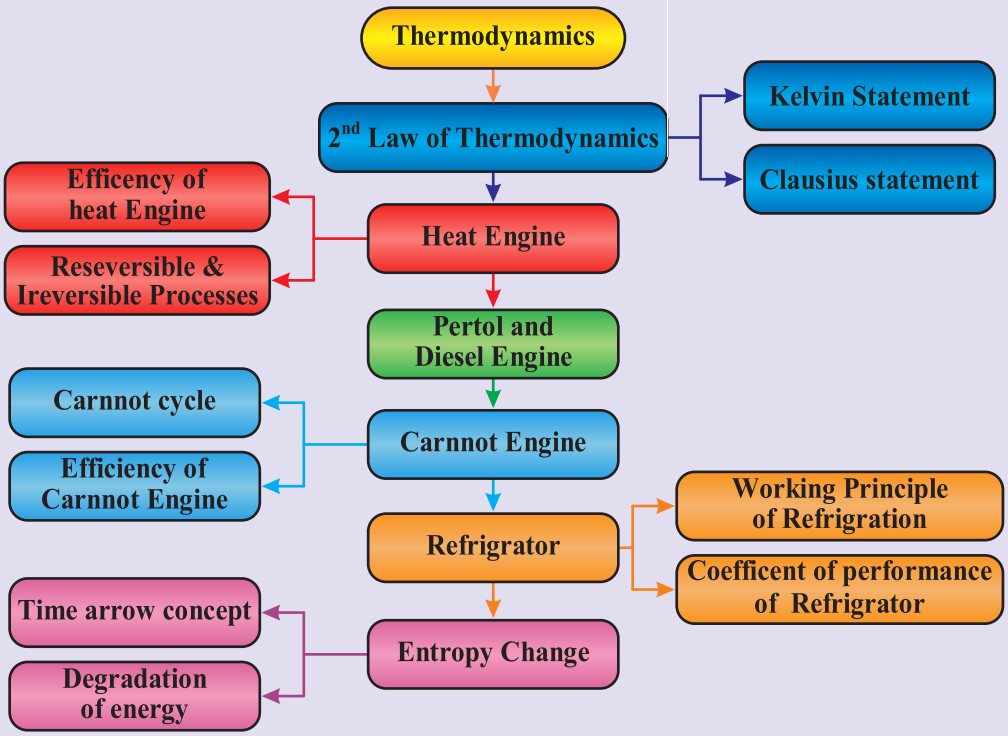
Choose the correct answer:

1. What type of process occurs when a system exchanges both heat and work with its surroundings, and there is no change in internal energy?
   (a) Isothermal process (b) Adiabatic process
   (c) Isobaric process (d) Isochoric process
2. During an isobaric process, what remains constant?
   (a) Temperature (b) Pressure (c) Volume (d) Internal energy
3. In which thermodynamic process does a system exchange heat with its surroundings but undergoes no change in temperature?
   (a) Isothermal process (b) Adiabatic process
   (c) Isobaric process (d) Isochoric process
4. What is the characteristic of an adiabatic process?
   (a) Constant pressure (b) No heat exchange with the surroundings
   (c) Constant temperature (d) Constant volume
5. In an isochoric process, what is the primary feature?
   (a) Constant temperature (b) No work done
   (c) Constant volume (d) No heat exchange
6. What is internal energy in a thermodynamic system?
   (a) The energy associated with motion
   (b) The energy associated with the system's position
   (c) The sum of kinetic and potential energy
   (d) The total energy contained within the system
7. How is the change in internal energy (ΔU) defined in terms of heat (ΔQ and work (ΔW)
   (a) $\Delta U = \Delta Q + \Delta W$ (b) $\Delta U = \Delta Q - \Delta W$
   (c) $\Delta U = \Delta Q$ times W (d) $\Delta U = \Delta Q / \Delta W$
8. What is the internal energy of an ideal gas related to?
   (a) Temperature only (b) Pressure only
   (c) Volume only (d) Temperature, pressure, and volume
9. During an adiabatic process, what happens to the internal energy of the system?
   (a) Increases (b) Decreases
   (c) Remains constant (d) Depends on the specific heat
10. What is the equation for the internal energy change of a system in an isochoric process?
    (a) $\Delta U = \Delta Q + \Delta W$ (b) $\Delta U = \Delta Q - \Delta W$
    (c) $\Delta U = \Delta Q$ times ΔW (d) $\Delta U = \Delta Q / \Delta W$

Section (B): CRQs (Short Answered Questions):

1. Explain the concept of the first law of thermodynamics in your own words.
2. How does the first law of thermodynamics relate to the conservation of energy?

CONCEPT MAP



EXERCISE

Section (A): **Multiple Choice Questions (MCQs)**
Choose the correct answer:

1. A frictionless heat engine can be 100% efficient only if its exhaust temperature is:
   (a) 0°C (b) Equal to its input temperature
   (c) 0 K (d) half of its input temperature
2. A refrigerator, with its door open. The temperature of the room will
   (a) Rise (b) Fall
   (c) Remains the same (d) Rise or fall depending on the area of the room
3. Which device is not used in a diesel engine
   (a) Outlet Valve (b) Piston
   (c) Sparking Plug (d) Injector

4. Which one of the following processes is irreversible?
   (a) Slow compression of an elastic spring
   (b) Slow evaporation of a substance in an isolated vessel
   (c) Slow compression of a gas
   (d) A chemical explosion
5. According to 2nd law of thermodynamics, 100% conversion of heat into mechanical work is:
   (a) Possible (b) Possible, if the conditions are ideal
   (c) Not possible (d) Possible, if the process is adiabatic
6. The change in entropy is given by
   (a) $\Delta S = \frac{\Delta Q}{T}$ (b) $\Delta S = \Delta Q. T$ (c) $\Delta S = \frac{\Delta U}{T}$ (d) $\Delta S = \frac{\Delta W}{T}$
7. The net change in entropy as a system in a natural process is
   (a) Positive (b) Negative (c) Zero (d) Infinite
8. The efficiency of diesel engine is
   (a) Greater than petrol engine (b) Less than petrol engine
   (c) Equal to petrol engine (d) both have efficiency 1
9. Second law of thermodynamics states that:
   (a) Energy can't be converted. (b) Entropy decreases over time.
   (c) Heat flows from cold to hot. (d) Entropy increases over time
10. The process violates the 2nd law of thermodynamics is:
    (a) Refrigerator cooling. (b) Heat engine working.
    (c) Gases mixing. (d) Heat flowing from cold to hot.

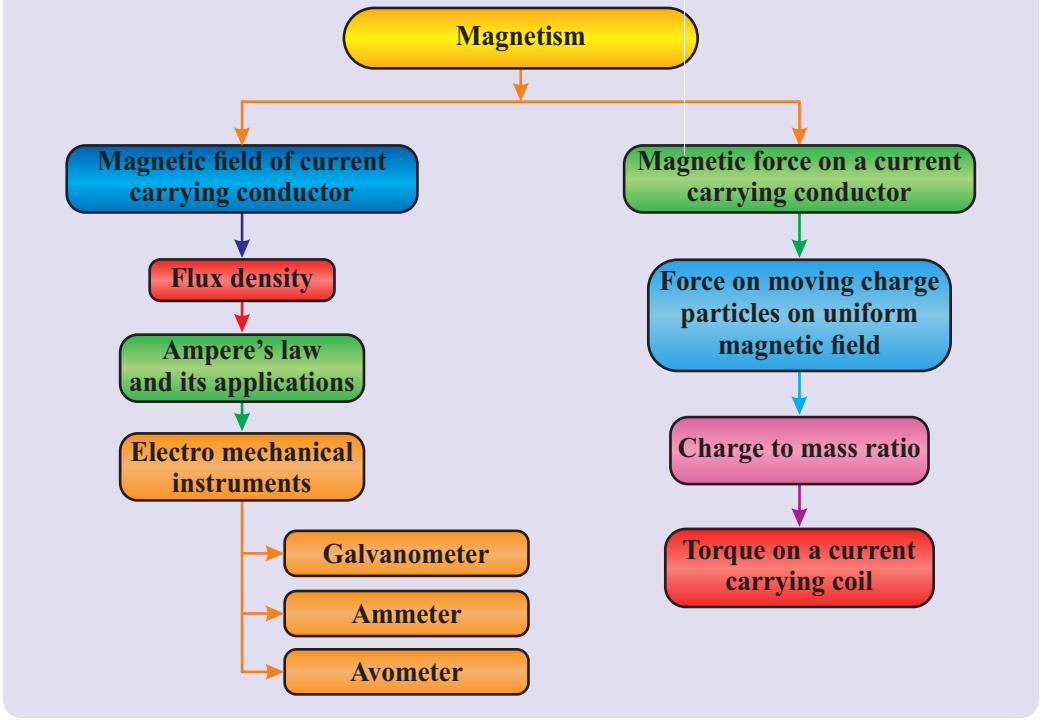
Section (B): CRQs (Short Answered Questions):

1. What are some factors that affect the efficiency of automobile engines?
2. What happens to the temperature of a room in which an air conditioner is left running on table in the middle of the room?
3. Under what conditions can heat be added to a system without changing its temperature?
4. Is it possible to cool a room by keeping the refrigerator door open?
5. When does the entropy of a system decrease?
6. Is it possible, according to the 2nd law of thermodynamics to construct an engine that is free from thermal pollution?
7. Can heat be completely converted to work?
8. Define that why entropy has often been called as “time arrow”.

Section (C): ERQs (Long Answered Questions):

1. Give the two statements of the second law of the thermodynamics. Elaborate the concept of entropy and state the second law of thermodynamics in terms of this concept.
2. Explain the working principle of heat engine and also derive the formula for its efficiency.
3. Describe the concept of reversible and irreversible process.
4. What is Carnot engine? Give the operation of Carnot cycle and show that the efficiency of even this engine is less than 100%.

CONCEPT MAP



EXERCISE

Section (A): **Multiple Choice Questions (MCQs)**

Choose the correct answer:

1. If were verse the direction of electric current, then the direction of magnetic field will be:
   (a) Same (b) reversed
   (c) tangent (d) normal
2. The application of magnetic field is:
   (a) microwave oven (b) magnetic levitation trains
   (c) electrolysis (d) plant photosynthesis
3. The equation $F=BIL$ can only be used, if the magnetic field, length of conductor, and electric current are:
   (a) At right angles to each other (b) in same direction
   (c) Anti-parallel to each other (d) anti-perpendicular to each other

4. The charge on a particle is doubled, and its velocity remains the same, how then the magnetic force on the particle will be:
   (a) doubled (b) halved
   (c) Is the same. (d) quadrupled
5. Strength of magnetic field of solenoid can be increased by:
   (a) Increasing number of turns
   (b) decreasing number of turns
   (c) increasing the current through the solenoid
   (d) inserting a ferromagnetic core (e.g., iron) into the solenoid
6. The magnetic field inside along solenoid is:
   (a) equal to zero
   (b) uniform
   (c) decreases as we go away from the center to surface
   (d) increases as we go to wards the surface
7. The force between two current carrying conductor arises due to
   (a) Magnetic effect of current (b) Polarization
   (c) Electromagnetic induction (d) Electrostatic interaction
8. To measure a higher voltage, what should you do with the voltmeter's internal resistance?
   (a) Increase it (b) Decrease it
   (c) Keep it the same (d) It doesn't affect the measurement
9. A proton moves perpendicular to a uniform magnetic field. What is the direction of the force experienced by the proton?
   (a) Parallel to the magnetic field (b) In the direction of the proton's velocity
   (c) Perpendicular to the magnetic field and the proton's velocity
   (d) Opposite to the direction of the proton's velocity
10. An electron moves parallel to a uniform magnetic field. What is the magnitude of the force experienced by the electron?
    (a) Maximum, since the electron is moving in the same direction as the field
    (b) Minimum, since the electron is moving perpendicular to the field
    (c) Zero, since the electron is moving parallel to the field
    (d) Depends on the speed of the electron

Section (B): CRQs (Short Answered Questions):

1. Charge particles are fired in vacuum tube hit a fluorescence screen. Will it be possible to know whether they positive or negative?
2. What is a solenoid, and how does it differ from a simple coil of wire?
3. Can a solenoid generate a magnetic field without any current flowing through it? Why or why not?
4. Explain why a toroid is often preferred over a straight solenoid when designing certain types of electrical components.
5. What role does the Ampere's circuital law play in understanding them agnetic field inside a toroid?
6. What is a galvanometer, and what is its primary function in an electrical circuit?



Section (A): **Multiple Choice Questions (MCQs)**
Choose the correct answer:

1. Electricity production by magnetism involves
   (a) the conversion of chemical energy to electrical energy.
   (b) the use of static electricity for power generation.
   (c) the movement of conductors within magnetic fields.
   (d) the absorption of solar energy by photovoltaic cells.
2. Induced e.m.f can be generated by relative movement, known as the generator effect, and by changing magnetic fields, known as the transformer effect.
   (a) True in transformers only.
   (b) False in all cases.
   (c) Applicable to both cases.
   (d) True only for static magnetic fields.
3. The magnitude of induced e.m.f. increases with
   (a) the decrease in the speed of the conductor's motion.
   (b) the decrease in the strength of the magnetic field.
   (c) the increase in the number of turns in the coil.
   (d) the reduction in the area of the coil.
4. Faraday's law of electromagnetic induction states
   (a) that current in a circuit always opposes the change in magnetic flux.
   (b) that the induced e.m.f. is directly proportional to the rate of change of magnetic flux.
   (c) that a changing electric field produces a magnetic field.
   (d) that magnetic fields can only be produced by electric currents.
5. Lenz's law predicts the direction of an induced current
   (a) to conserve electric charge.
   (b) by stating that the induced current will oppose the change causing it.
   (c) by following the direction of the applied magnetic field.
   (d) based on the magnitude of the electric field.
6. Eddy currents produce:
   (a) static electricity in conductive materials.
   (b) alternating magnetic fields with no heating effects.
   (c) magnetic fields that oppose the inducing field and cause heating in conductors.
   (d) uniform electric fields that have no impact on power generation.
7. Laminated iron cores are used in electric motors and transformers to
   (a) reduce the weight of the device.
   (b) enhance the mechanical strength.
   (c) minimize eddy current losses and reduce heating.
   (d) increase the magnetic permeability of the core.

8. Self-induction is the phenomenon where
   (a) a changing electric field induces a magnetic field.
   (b) a changing magnetic field within a coil induces an emf in the same coil.
   (c) a constant magnetic field induces a constant emf.
   (d) two coils induce emf in each other through mutual induction.
9. An inductor stores electric potential energy in:
   (a) the electric field around it.
   (b) the magnetic field within its coil.
   (c) the capacitance of its windings.
   (d) the heat generated by its resistance.
10. Transformers work on the principle of
   (a) converting direct current (DC) to alternating current (AC).
   (b) electromagnetic induction between primary and secondary coils.
   (c) generating electricity through chemical reactions.
   (d) using permanent magnets to maintain a constant voltage.

Section (B): CRQs (Short Answered Questions):

1. Explain Faraday's law of electromagnetic induction.
2. State Lenz's law and discuss its significance in the context of electromagnetic induction.
3. How does a step-up transformer differ from a step-down transformer?
4. Describe the principle of operation of an AC generator.
5. Define self-induction and explain how it occurs in a coil.
6. Define mutual induction and provide an example of a system exhibiting mutual induction.
7. How does the arrangement of coils influence the degree of mutual induction between them?
8. Define motional electromotive force (emf)
9. Differentiate between motional emf and electromagnetic induction in terms of their fundamental principles.
10. Differentiate between AC (alternating current) and DC (direct current) in the context of long-distance power transmission.

Section (C): ERQs (Long Answered Questions):

1. How does the strength of the magnetic field affect the induced electromotive force (emf) in a coil according to Faraday's law? b. Explain the relationship between the magnetic field strength and the magnitude of the induced emf.
2. In Faraday's law, why is the area of the coil an important factor in determining the induced emf?
3. How does changing the orientation of a coil with respect to the magnetic field influence the induced emf? Also describe the role of the number of turns in a coil in the context of induced emf. How does increasing the number of turns in a coil affect the induced emf, and why?
4. According to Faraday's law, why is the rate of change of magnetic flux crucial for inducing emf? Discuss how the speed of motion or change in magnetic field influences the induced emf.

EXERCISE

Section (A): **Multiple Choice Questions (MCQs)**

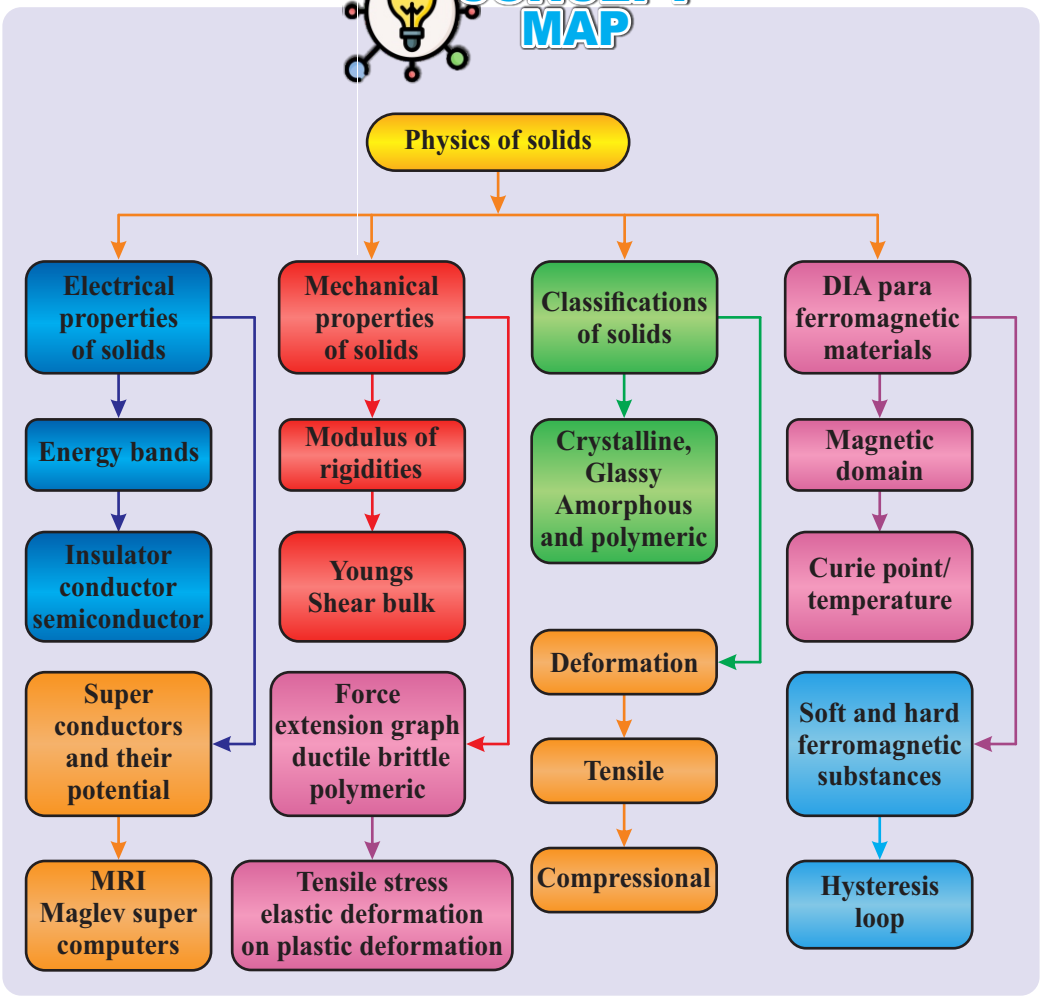
Choose the correct answer:

1. In alternating current, the flow of electric charge periodically reverses direction. The term for this specific phenomenon is:
   (a) Alternation (b) Oscillation (c) Cycle (d) Resistance
2. The frequency of standard household alternating current in Pakistan is:
   (a) 50 Hz (b) 60 Hz (c) 100 Hz (d) 120 Hz
3. The parameter for measuring the rate of change of alternating current or voltage with respect to time is:
   (a) Amplitude (b) Frequency (c) Phase angle (d) Peak Voltage
4. In AC circuits, what is the phase difference between current and voltage in a purely resistive circuit?
   (a) 0^0 (b) 45^0 (c) 90^0 (d) 180^0
5. A device used to measure the root mean square (RMS) value of an AC voltage or current is:
   (a) Voltmeter (b) Oscilloscope (b) Ammeter (d) AVO meter
6. In an AC circuit. The term "reactance" represent the:
   (a) Ohmic resistance only
   (b) Resistance to DC currents only
   (c) Capacitive and inductive opposition to current flow
   (d) Total impedance, including resistance and phase angle
7. A type of a circuit characterized by equal values of resistance and reactance is:
   (a) Resistive (b) Inductive (c) Capacitive (d) Resonant
8. In an AC sine wave, the peak-to-peak voltage is:
   (a) Double the peak voltage
   (b) The difference between the peak voltage and zero
   (c) The sum of the peak voltage and zero
   (d) The difference between the positive and negative peak voltages
9. What is the power factor of a purely resistive circuit?
   (a) 0 (b) 0.5 (c) 1 (d) -1
10. The purpose of a transformer in an AC power distribution system.
    (a) To convert AC to DC (b) To step up or step down voltage
    (c) To store electrical energy (d) To regulate current flow

Section (B): **CRQs (Short Answered Questions):**

1. Define RMS voltage and explain its significance in AC circuits.
2. Explain the difference between peak voltage, RMS voltage, and average voltage in AC circuits.
3. Define alternating current (AC) and explain how it differs from direct current (DC).
4. Define reactance in AC circuits and differentiate between capacitive and inductive reactance.

CONCEPT MAP



EXERCISE

Section (A): **Multiple Choice Questions (MCQs)**
Choose the correct answer:

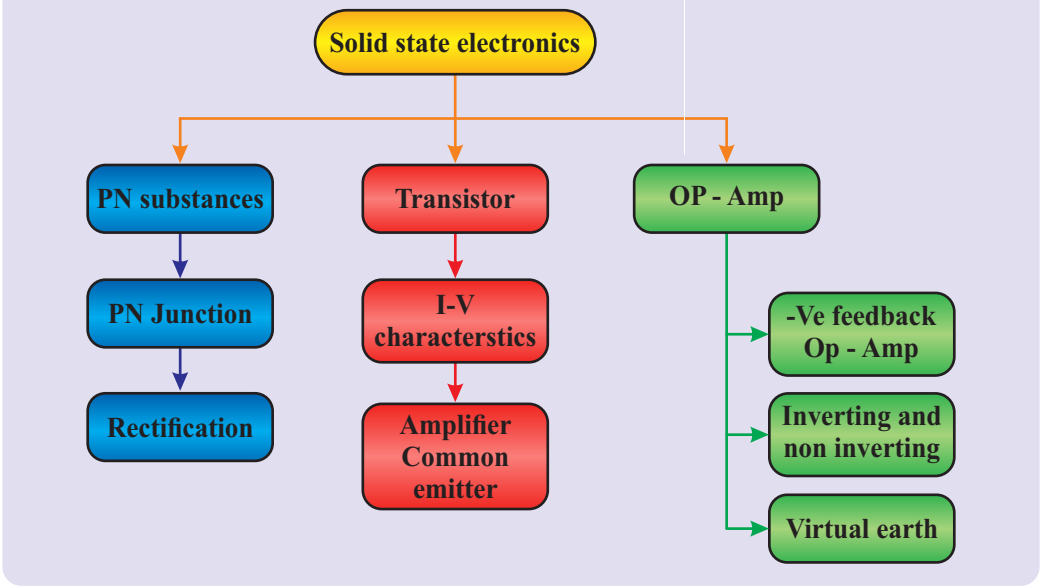
1. The property of a body by virtue of which it tends to regain its original size and shape when the applied force is removed is called
   (a) Elasticity (b) Plasticity (c) Rigidity (d) Compressibility
2. Substances which can be stretched to cause large strains are called
   (a) Brittle (b) Ductile (c) Plastic (d) Elastomer

3. If the load is increased beyond the point, the strain increases rapidly for even a small change in the stress.
   (a) Elastic point (b) Yield point (c) Plastic point (d) Fracture point
4. The reciprocal of the bulk modulus is called
   (a) Compressibility (b) Volume stress
   (c) Modulus of rigidity (d) Volume strain
5. Which of the following statements is/are wrong?
   i. Hollow shaft is much stronger than a solid of the same length and same mass.
   ii. Reciprocal of bulk modulus of elasticity is called compressibility.
   iii. It is difficult to twist along rod as compared to small rod.
   (a) III only (b) I only (c) II and III (d) I and II
6. Metals are good conductors of heat and electricity. This property is conferred by
   (a) Covalent (b) Ionic (c) Metallic (d) Hydrogen
7. For a metallic crystal, delocalized electrons occupied band is:
   (a) Conduction band (b) Valence band
   (c) Conduction and valence bands (d) there are no delocalized electrons
8. The semiconductors have resistivity
   (a) Between conductor and insulator (b) More than insulators
   (c) Less than conductors
   (d) Depending upon the semiconductor material property.
9. A ferromagnetic substance becomes a permanent magnet when it is placed in a magnetic field because
   (a) All the domains get oriented in the direction of the magnetic field.
   (b) All the domains get oriented in the direction opposite to the direction of the magnetic field.
   (c) Domains get oriented randomly.
   (d) Domains are not affected by the magnetic field.
10. Identify methods to demagnetize a ferromagnet.
    (a) By cooling, heating, or submerging in water
    (b) By heating, hammering, and spinning it in an external magnetic field
    (c) By hammering, heating, and rubbing with cloth
    (d) By cooling, submerging in water, or rubbing with cloth

Section (B): CRQs (Short Answered Questions):

1. Why are the springs made of steel and not of copper?
2. The breaking force for a wire is F. What will be the breaking force for two parallel wires of the same size?
3. Distinguish between intrinsic and extrinsic semiconductor.
4. A wire is replaced by another wire of same length and material but of twice the diameter. What will be the effect on the:
   (a) Increase in its length under a given load? (b) Maximum load which it can bear?
5. Sand does not possess any definite shape and volume, still it is solid. Give reason.
6. Specify the importance of stress-strain curve.
7. Why liquids don't possess rigidity?

CONCEPT MAP



EXERCISE

Section (A): **Multiple Choice Questions (MCQs)**
Choose the correct answer:

1. A semiconductor is an element with a valence electron _____.
   (a) Four (b) Eight (c) Two (d) One
2. A pure semiconductor is known as
   (a) Extrinsic (b) Intrinsic (c) Transistor (d) Diode
3. An acceptor atom is also called
   (a) Penta-valent atom (b) Trivalent atom
   (c) Minority carrier (d) Majority carrier
4. With which one of the following elements silicon should be doped so as to give p-type of semiconductor?
   (a) Germanium (b) Arsenic (c) Selenium (d) Boron
5. For a full-wave rectifier, the output frequency
   (a) Equals one-half the input frequency (b) Equals the line frequency
   (c) Equals two times the input (d) Is three times the line frequency

6. LED construction needs a semiconductor material is:
   (a) Silicon (b) Germanium (c) Gallium (d) Gallium arsenide
7. The frequency of a half-wave signal is
   (a) Twice the line frequency (b) Equal to the line frequency
   (c) One-half of the line frequency (d) One-fourth the line frequency
8. The voltage gain of an emitter follower circuit is
   (a) High (b) Low (c) Very high (d) Moderate
9. What is also called as the conventional amplifier?
   (a) Common-collector circuit (b) Emitter follower circuit
   (c) Common base circuit (d) Common emitter circuit
10. An op-amp with negative feedback provides _____ output parameter.
   (a) Gain (b) Bandwidth
   (c) Input-output impedance (d) Amplified

Section (B): CRQs (Short Answered Questions):

1. Describe a *p-n* junction (diode) and how holes and electrons are produced in semiconductor?
2. Define and distinguish between p-n-p & n-p-n transistors?
3. Explain common-base and common collector configurations?
4. Describe the operations of transistors.
5. Explain the use of transistors as a switch and an amplifier (common-emitter).
6. How would you understand the effects of negative feedback on the gain of an operational amplifier?
7. Draw and briefly describe the circuit diagrams for both the inverting and the non-inverting amplifier for single signal input?
8. Derive an expression for the gain of inverting amplifiers by using virtual earth approximation.
9. Describe the properties of an ideal operational amplifier.

Section (C): ERQs (Long Answered Questions):

1. Define Intrinsic (pure) and doped semiconductors. And How the N-type and P-type semiconductors are produced
2. Describe a P-N junction (diode) discus its forward and reverse biasing.
3. Describe the I-V characteristic curves of P-N junction.
4. Define rectification and describe the use of diodes for half and full wave rectifications.
5. Describe the function and use of LED, Photodiode and Photo voltaic cell Describe the function and use of LED, Photodiode and Photo voltaic cell.
6. Explain the use of transistors as a switch and an amplifier (common-emitter).
7. Describe the properties of an ideal operational amplifier.

EXERCISE

Section (A): Multiple Choice Questions (MCQs)

Choose the correct answer:

1. The voltage level, typically associated with a 'low' state in digital electronic is:
   (a) -5 volts (b) 0 volts (c) +5 volts (d) +10 volts
2. In binary representation, the correspond of '0' in digital electronics is:
   (a) Low state (b) High state (c) Open circuit (d) Closed circuit
3. The primary purpose of a truth table in digital electronics is to:
   (a) Determine input voltage
   (b) Measure circuit resistance
   (c) Analyze output states based on input combinations
   (d) Calculate circuit power
4. Which logic gate combination produces an output of '1' when at least one input is '1'?
   (a) AND (b) OR (c) XOR (d) NOT
5. In a circuit with a 2-input OR gate, the status of the lamp if either switch is closed
   (a) OFF (b) ON (c) Blinking (d) Flickering
6. The behavior of 2 input AND gate, when both switches are closed, Lamp is
   (a) OFF (b) ON (c) blinking (d) flickering
7. In a 3-input OR gate, if all inputs are '0'sothe output will be:
   (a) 1 (b) 0 (c) Undefined (d) Both A and B
8. If the input is '1', The output of a NOT gate will be:
   (a) 1 (b) 0 (c) Undefined (d) Both A and B
9. In a 2-input XOR gate, when both inputs are '1' than the output will be
   (a) 1 (b) 0 (c) Undefined (d) Both A and B
10. Which combination of logic gates is equivalent to an XOR gate?
    (a) AND + OR (b) NOT + NOR (c) OR + NAND (d) AND + NAND

Section (B): CRQs (Short Answered Questions):

1. Explain the significance of signal levels in digital electronics and how they are represented in terms of voltage.
2. Describe the binary representation used in digital electronics. How is this representation related to the 'high' and 'low' states?
3. Choose any three logic gate symbols and explain their operations.
4. Create a combination of logic gates that mimics the behavior of an XOR gate.

Section (C): ERQs (Long Answered Questions):

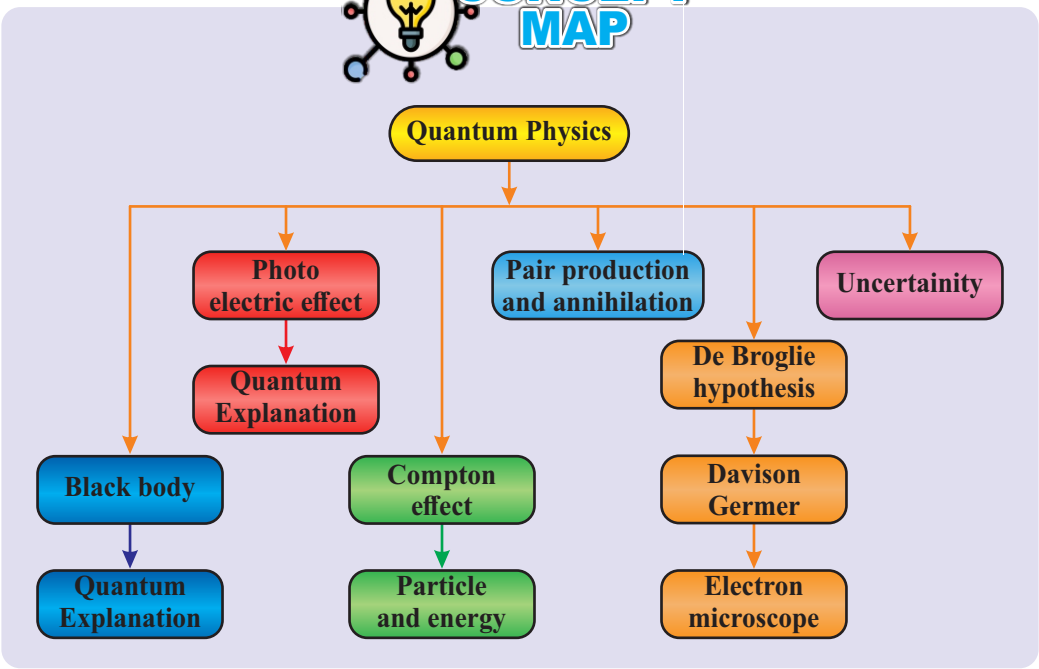
1. Design a simple circuit using a 2-input AND gate, a switch, a lamp, and a battery. Explain how the circuit operates and the conditions under which the lamp will be illuminated.
2. Construct a truth table for a 3-input OR gate. Explain how the truth table represents the relationship between input combinations and the resulting output.



Section (A): **Multiple Choice Questions (MCQs)**
Choose the correct answer:

1. The General Theory of Relativity was a new way of understanding of
   (a) The speed of light (b) gravity
   (c) mass (d) force
2. An object at rest has a mass of 1 kg. What is its mass when it is moving at a speed of 0.9c?
   (a) Infinite (b) 1.2 kg (c) 2.3 kg (d) 1 kg
3. The equivalence of principle in general relativity is:
   (a) The equivalence of inertial and gravitational mass
   (b) The equivalence of electric and magnetic fields
   (c) The equivalence of space and time
   (d) The equivalence of matter and energy
4. Which of the following phenomena is NOT predicted by special relativity?
   (a) Time dilation (b) Length contraction
   (c) Gravitational waves (d) Relativistic mass increase
5. What does the equivalence principle state about acceleration and gravity?
   (a) They are completely different forces (b) They are indistinguishable for an observer
   (c) Gravity is stronger than acceleration (d) Acceleration cancels out gravity
6. Light passing near a massive object like a star will bend due to:
   (a) Gravitational lensing (b) refraction
   (c) reflection (d) diffraction
7. What is the main reason astronomers cannot directly observe black holes?
   (a) They are too small (b) They deflect light
   (c) They are too far away (d) their immense gravity traps light
8. Objects cannot exceed the speed of light because:
   (a) Their mass becomes infinite (b) Their length becomes zero
   (c) Time slows down to zero (d) they lose all energy
9. Why is the Galilean transformation not valid at high speeds?
   (a) It cannot handle accelerating frames
   (b) It violates the constancy of the speed of light
   (c) It only works in flat spacetime
   (d) It neglects time dilation effects
10. The example of inertial frame in our everyday life is:
    (a) A car accelerating on a highway
    (b) A train moving smoothly at constant speed
    (c) A person standing on a spinning platform
    (d) an airplane encountering turbulence

CONCEPT MAP



EXERCISE

Section (A): **Multiple Choice Questions (MCQs)**
Choose the correct answer:

1. If h is the Planck constant, then $\hbar$ is:
   (a) $2\pi h$ (b) $2h$ (c) $h/2$ (d) $h/2\pi$ (e) $2h/\pi$
2. In a photoelectric effect experiment the stopping potential is:
   (a) the energy required to remove an electron from the sample
   (b) the kinetic energy of the most energetic electron ejected
   (c) the potential energy of the most energetic electron ejected
   (d) the electric potential that causes the electron current to vanish
3. The work function for a certain sample is 2.3 eV. The stopping potential for electrons ejected from the sample by 7.0×10^{14}-Hz electromagnetic radiation is:
   (a) 0 (b) 0.60 V (c) 2.3V (d) 2.9V (e) 5.2V
4. In Compton scattering from stationary particles the maximum change in wavelength can be made smaller by using:
   (a) higher frequency radiation (b) lower frequency radiation
   (c) more massive particles (d) less massive particles

5. Of the following, Compton scattering from electrons is most easily observed for:
   (a) infrared light (b) visible light
   (c) ultraviolet light (d) X-rays
6. In the Compton scattering from stationary electrons the largest change in wavelength occurs when the photon is scattered through
   (a).0° (b) 45° (c) 90° (d) 180°
7. A free electron has a momentum of 5.0×10^{-24} kg · m/s. The wavelength of its wave function is:
   (a) 1.3×10^{-8}m (b) 1.3×10^{-10} m
   (c) 2.1×10^{-11} m (d) 2.1×10^{-13} m
8. In Photoelectric Effect, when light fall on the surface of metal, the material should emit
   (a) Electrons (b) Protons
   (c) Positrons (d) Neutrons
9. Threshold frequency is defined as the ..... frequency of incident light which can cause photo electric emission
   (a) Maximum (b) Minimum
   (c) Average (d) highest
10. The amount of energy which is necessary to start photo electric emission is called:
   (a) Maximum (b) Average
   (c) Minimum (d) Littlest

Section (B): CRQs (Short Answered Questions):

1. Differentiate between wave and particle.
2. Is it possible for the de Broglie wavelength of a particle?
3. Estimated Broglie wavelength of a cricket ball on the pitch?
4. Differentiate between the continuous and discrete emission of radiation?
5. What is threshold frequency?
6. How has the photoelectric effect been applied in real-world technologies or devices, and what are its practical implications?
7. What are the advantages of an electron microscope over an optical microscope?
8. Is it possible to create only an electron through matter and photon interaction?
9. Give construction of electron microscope?
10. Elaborate the particle nature of electromagnetic radiation.

EXERCISE

Section (A): Multiple Choice Questions (MCQs)
Choose the correct answer:

1. Atomic spectra are also known as:
   (a) Discrete spectra (b) Line spectra
   (c) Emission spectrum (d) Continuous spectra
2. The ratio of kinetic energy to total energy for an electron in a Bohr orbit is:
   (a) 1:1 (b) 2:3 (c) 1:2 (d) 2:1
3. The Bohr radius increases as the principal quantum number:
   (a) increases (b) decreases
   (c) remains constant (d) oscillates
4. Laser beams consist of:
   (a) Highly coherent electrons (b) Highly coherent photons
   (c) Highly coherent phonons (d) Highly coherent neutrons
5. The ruby laser is an example of:
   (a) Optical pumping (b) Electrical pumping
   (c) Chemical pumping (d) Thermal pumping
6. Laser action requires a medium with at least:
   (a) Three energy levels (b) Four energy levels
   (c) Two energy levels (d) Five energy levels
7. Population inversion occurs in:
   (a) Active medium (b) Passive medium
   (c) Gaseous medium (d) Vapour medium
8. X-rays transfer _____ to metals.
   (a) Energy (b) Force (c) Pressure (d) Momentum
9. X-rays are deflected by:
   (a) Magnetic fields (b) Electric fields
   (c) Gravitational fields (d) No fields
10. Doubling the voltage of an X-ray tube:
    (a) Halves the intensity (b) Keeps the intensity unchanged
    (c) Doubles the intensity (d) Quadruples the intensity

Section (B): CRQs (Short Answered Questions):

1. Why do different elements have different spectra?
2. In the Bohr model, how many times larger is the radius of the fifth Bohr orbit compared to that of the first Bohr orbit?
3. What is the difference between X-Rays and Gamma Rays?
4. Sate the properties of X-Rays, which makes it possible to detect cracks in bones.

EXERCISE

Section (A): Multiple Choice Questions (MCQs)

Choose the correct answer:

1. Identify particle x in the following nuclear reaction ${}^{9}_{4}Be + {}^{4}_{2}He \rightarrow {}^{12}_{6}C + x$
   (a) Electron (b) proton (c) neutron (d) photon
2. In the equation ${}^{27}_{13}Al + {}^{4}_{2}He \rightarrow {}^{30}_{15}P + X$,The correct symbol for X is:
   (a) ${}^{-1}_{0}e$ (b) ${}^{1}_{1}H$ (c) ${}^{4}_{2}He$ (d) ${}^{1}_{0}n$
3. Beta rays emitted by a radioactive material are:
   (a) Electromagnetic radiations (b) The electrons orbiting around the nucleus
   (c) Charged particles emitted by the nucleus (d) Neutral particles
4. The number of α and β particles emitted in the following radioactive decay is:
   ${}^{200}_{90}X \rightarrow {}^{168}_{80}Y + {}^{?}_{?}X$
   (a) 8 and 6 (b) 6 and 8 (c) 8 and 8 (d) 6 and 6
5. If radium has half-life of 5 years. Thus for a nucleus in a sample of radium, the probability of decay in ten years is:
   (a) 50% (b) 75% (c) 100% (d) 60%
6. Emission of β particles by an element affects its mass number **A** in the following way:
   (a) Increases by 1 (b) decreases by 1
   (c) Increases by 2 (d) remains the same
7. As the number of nucleons in a nucleus increases, the binding energy per nucleon:
   (a) Increases continuously with mass number
   (b) Decreases continuously with mass number
   (c) remains constant with mass number
   (d) First increases and then decreases with increase of mass number
8. Moderator in a nuclear reactor slows down the neutrons to:
   (a) Decrease the probability of escape (b) Increase the probability of nuclear fission
   (c) Decrease the probability of absorption (d) all of the above
9. Emission of β^{+}particles by an element affects its atomic number Z in the following way:
   (a) Increases by 1 (b) decreases by 1
   (c) Increases by 2 (d) remains the same
10. The half life period of a radioactive element is 100 days. After 400 days, 16 g of the element will be reduced to
    (a) 8g (b) 4g (c) 2g (d) 1g

Section (B): CRQs (Short Answered Questions):

1. Why are protons and neutrons necessary for the stability of an atomic nucleus?
2. How do isotopes of an element differ and why are these differences significant?
3. Name the two methods of controlling a chain reaction.
4. Why is nuclear decay described as spontaneous and random?
5. Explain what happens during the nuclear fusion process.



Section (A): **Multiple Choice Questions (MCQs)**
Choose the correct answer:

1. The Standard Model classifies elementary particles into two main groups:
   (a) Baryons and Leptons (b) Fermions and Bosons
   (c) Quarks and Gluons (d) Hadrons and Mesons
2. The following is NOT a flavor of quark:
   (a) Up (b) Down (c) Electron (d) Top
3. The charge of an up quark is:
   (a) +1/2e (b) -1/2 e (c) +2/3e (d) -2/3e
4. The concept of "quark confinement" implies that:
   (a) Quarks cannot exist as free particles outside of hadrons.
   (b) Quarks are always found in pairs with opposite charges.
   (c) Quarks have a strong affinity for gluons.
   (d) Quarks are the fundamental building blocks of all matter.
5. The primary role of the Higgs boson in the Standard Model is:
   (a) Mediating electromagnetic interactions (b) Providing mass to other particles
   (c) Transmitting the strong nuclear force (d) Creating dark matter
6. According to the Standard Model, the term "color" refers to:
   (a) Visible light spectrum (b) Charge property of quarks
   (c) Mass of particles (d) Spin of particles
7. A particle made up of a quark and an antiquark is called:
   (a) Lepton (b) Baryon
   (c) Meson (d) Neutrino
8. An elementary particle that feels all three fundamental forces (electromagnetic, weak, and strong nuclear forces) is:
   (a) Lepton (b) Quark
   (c) Electron (d) Neutrino
9. The primary function of a Geiger-Müller counter in particle physics is to:
   (a) Measure the velocity of particles
   (b) Detect and count ionizing radiation
   (c) Create antimatter particles
   (d) Generate magnetic fields
10. In a Geiger-Müller counter, the gas commonly used to detect ionizing radiation is:
    (a) Oxygen (b) Neon (c) Argon (d) Helium